

PROPERTY LISTING

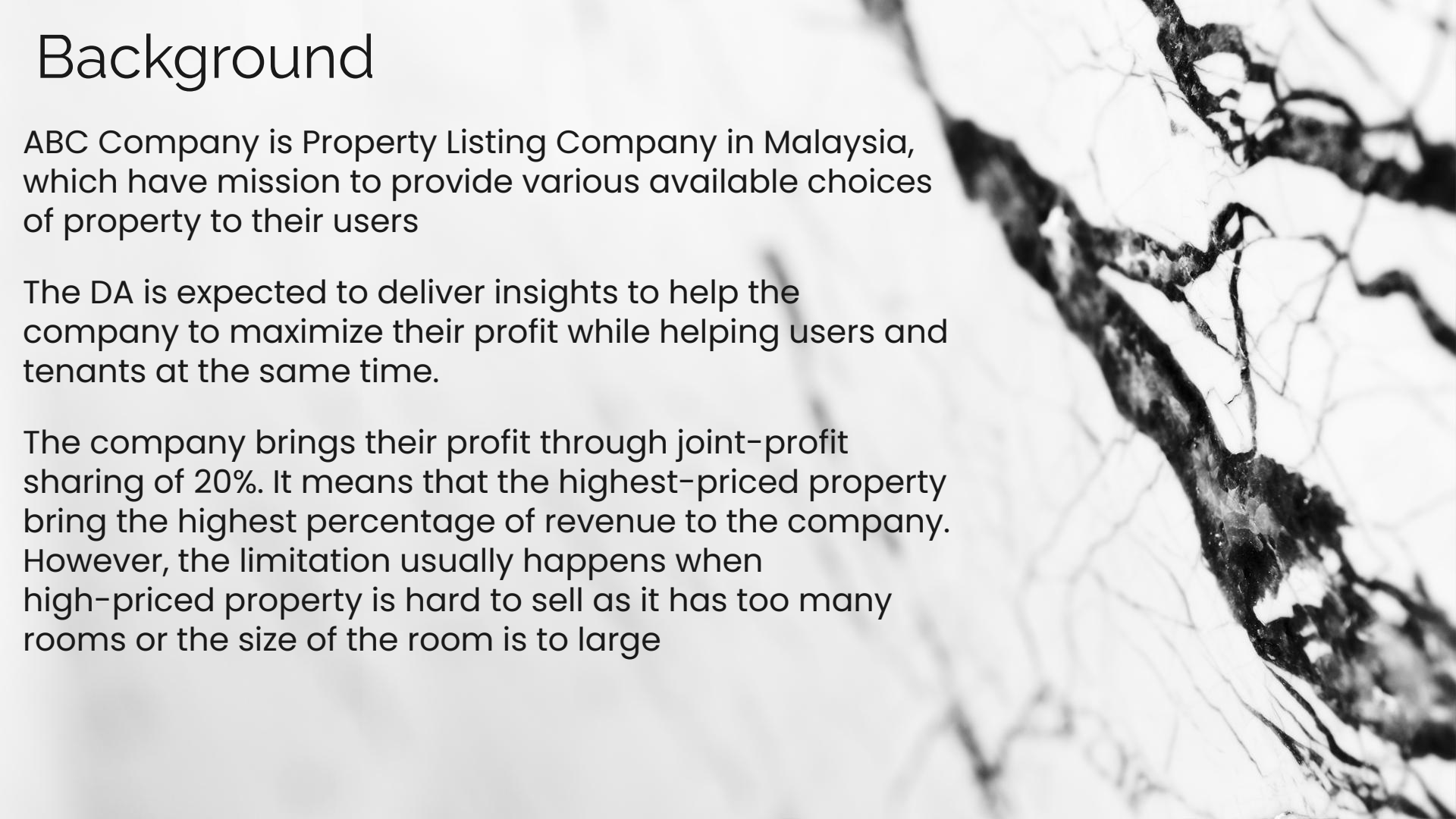
By Akhmad Fauzi Nurulhamzah

Background

ABC Company is Property Listing Company in Malaysia, which have mission to provide various available choices of property to their users

The DA is expected to deliver insights to help the company to maximize their profit while helping users and tenants at the same time.

The company brings their profit through joint-profit sharing of 20%. It means that the highest-priced property bring the highest percentage of revenue to the company. However, the limitation usually happens when high-priced property is hard to sell as it has too many rooms or the size of the room is to large



01

Data Cleaning

How we clean the dataset

02

Descriptive Statistics

You can describe the topic of the section here

03

EDA

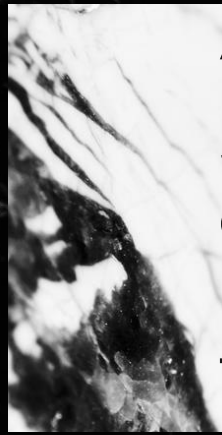
How we get Insight from data

04

Correlation & Regression

Describe the relationships between the factors that determine price. Additionally, use regression to forecast price.

Business Understanding



"How to identify the best way to maximize 20% sharing profit from selling the property by overcome the limitations that is hard to sell high price properties with too many rooms or too large"

"The data will help the company to find and choose the right properties to sell by analyzing what kind of property that give most profit this year"

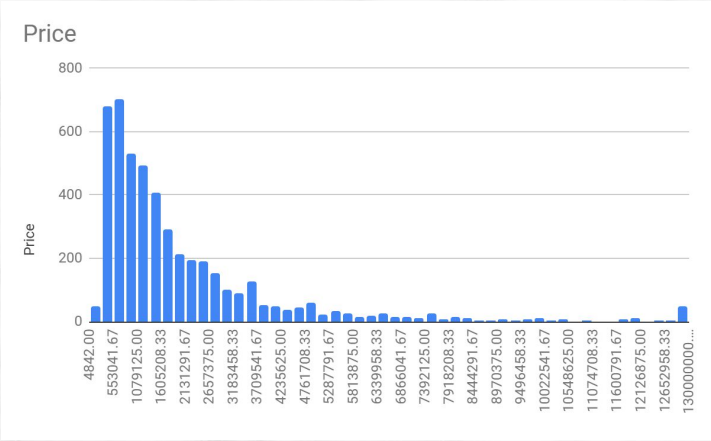
We can use this columns to extract insights using statistics and pivot table/using chart

Data Cleaning

No	Column	What to do	Remark/Reference	Reason
1	All	Checking extra spaces	Using "Trim Whitespace" Function	Extra space can make the data processing difficult to do, because it is also read as string
2	Location	Delete ' , Kuala Lumpur'	Using Find and Replace	Because it is unnecessary data
3	Location	Correct ADIVA-Desa Park City	Using Filter and Replace	Filter the data to detect inconsistency typing and format
4	Location	Correct SEMARAK to Semarak	Using Filter and Replace	Filter the data to detect inconsistency typing and format
5	Price	Delete ' , RM'	Using Find and Replace	Because we need to calculate the price and size so we need it to be number
6	Rooms	Replace Studio with 1	Using Find and Replace	in order to calculate this colum we need the data format to be in number format
7	Rooms	Edit 3+1, 2+1, 5+1 (uniformize a+b	Using Split Text to Columns and sum	we can't calculate data in this format
8	Property Type	Remove unneccesarry data	Using Split Text to Columns	To make the column easier to analyze
9	Property Type	Delete '/Link House'	Using Find and Replace	redundant, unneccesarry data
10	Property Character	Correct Build Up: to Build Up	Using Find and Replace	To make the column easier to analyze because the data is consisten

No	Column	What to do	Remark/Reference	Reason
11	Property Character	clear sq.m.	Using Filter and Delete	to make tge grouping right
12	Size	Split Number and unit	Using Split Text to Columns	To make the column easier to analyze/calculate
13	Size	Convert data Acre and Sq.m to Sqft	Using multiplication with sq.feat coversion	In aggregation, the same units of measurement are required
14	Size	Corret a x b	Using Split Text to Columns and multiplication	In aggregation, the same units of measurement are required
15	Size	Correct 22’x100&#8217 22×100	Using Split Text to Columns and multiplication	In aggregation, the same units of measurement are required
16	Size	Clear typo and false data (850sf~1000sf)	Using Filter and Delete	In aggregation, the same units of measurement are required
17	Furnishing	Replace Blank with Unknown	Using Sort, find and Replace	because uniformity is important
18	Size	Delete Blank and 0	Using Sort, find and Delete	Because we cant analise blank or 0 size
19	Price	Delete Blank	Using Sort, find and Delete	Because we cant analise blank prize
20	Price per sqft	Make Prrice per sqft Column	By devide price with size	to know whic propery has the most value per sqft
21	Price	Edit typo data	Edit 408 to 408.000	it is reasonable to edit the price by looking property in the same character
22	Size	Remove ilogical data	Filter Delete	it is not logical to have 17 and 790,000 sq.ft
23	All Number Format	Format all Number to Number	Format Number	Make sure every number is in the right format

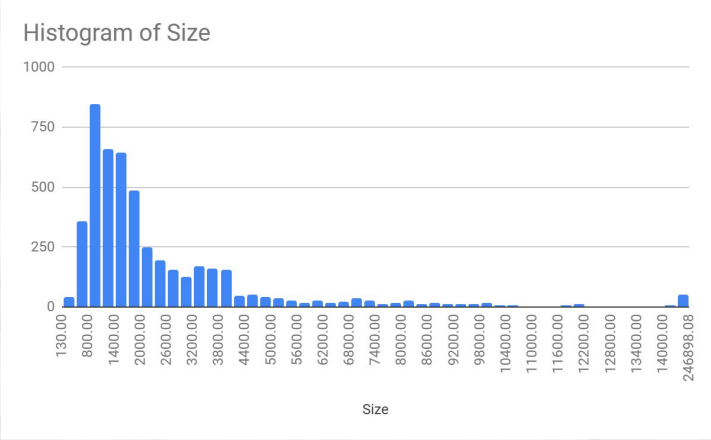
Deskriptif Analytics



Price	
Mean	2,166,614
Median	1,300,000.00
Mode	1,200,000.00
Standard Deviation	3,435,719
Skewness	15.15
Minimum	105,000
Maximum	130,000,000.00
Count	4,842

Insight:

1. The skewness point is >1 that indicates it is Highly positive Skewness.
2. It tell us that there are outlier data un "Price" and we need to se wheather the outlier is valid data or false data/typo
3. We need to use Median instead of mean for central tendency

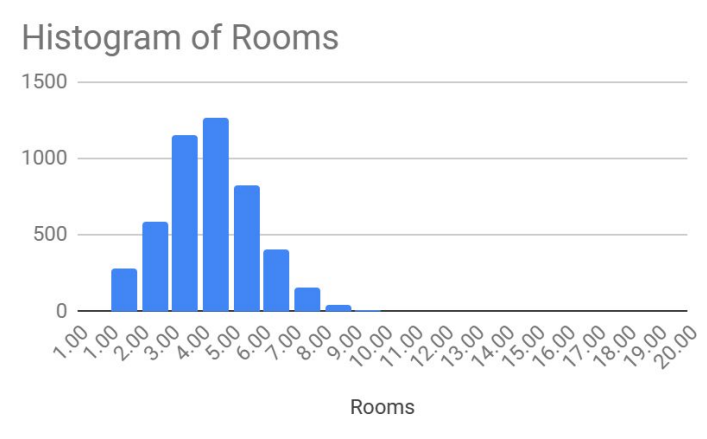


Size	
Mean	2,693.92
Median	1,650.00
Mode	1,650.00
Standard Deviation	5,899.89
Skewness	22.24
Minimum	130.00
Maximum	246,898.08
Count	4,842.00

Insight:

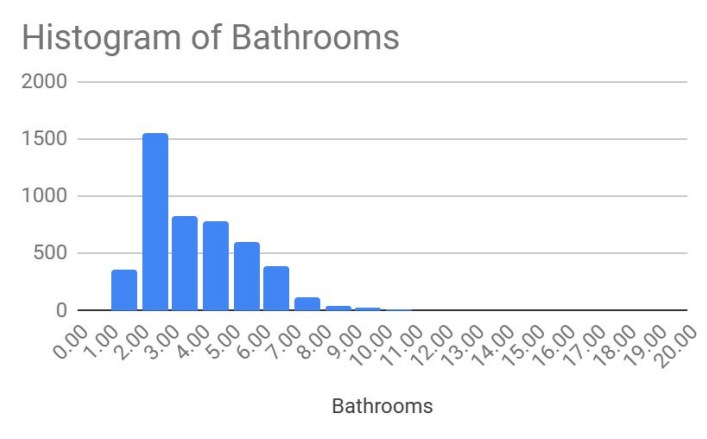
1. The skewness is highly positive.
2. even though we have remove the ilogical data, it still show us thats there are outlier that we need to check
3. We need to use Median instead of mean for central tendency

Deskriptif Analytics



Rooms	
Mean	4
Median	4
Mode	4
Standard Deviation	4
Skewness	0.65
Minimum	1
Maximum	20
Count	4736

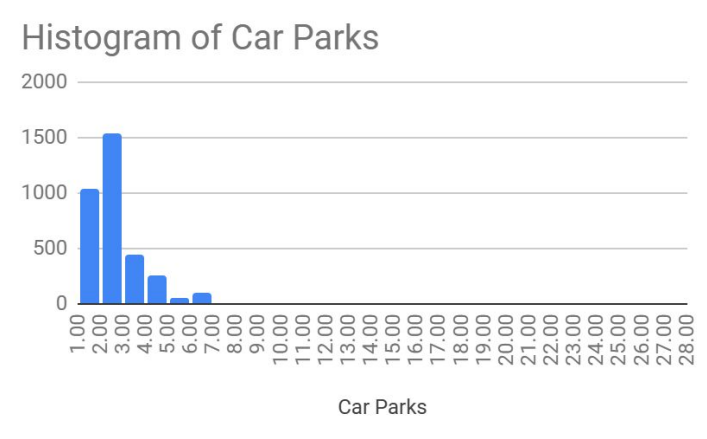
Insight:
1. the histogram is near normal distributed, there is positive skewness but slightly
2. Because the point is >0 and <1, it indicates trhat there are outlier but low
3. Thera are gaps (106) in count data that indicates not all properties have bathroom
4. We can still use mean or median, because for rooms the number is the same



Bathrooms	
Mean	3
Median	3.
Mode	2
Standard Deviation	2
Skewness	1.07
Range	19
Minimum	10
Maximum	20
Count	4,725

Insight:
1. the skewness is is positive
2. Because the point is <2, it indicates trhat there are outlier but not big
3. Thera are gaps (129) in count data that indicates not all properties have bathroom
4. We need to use Median instead of mean for central tendency

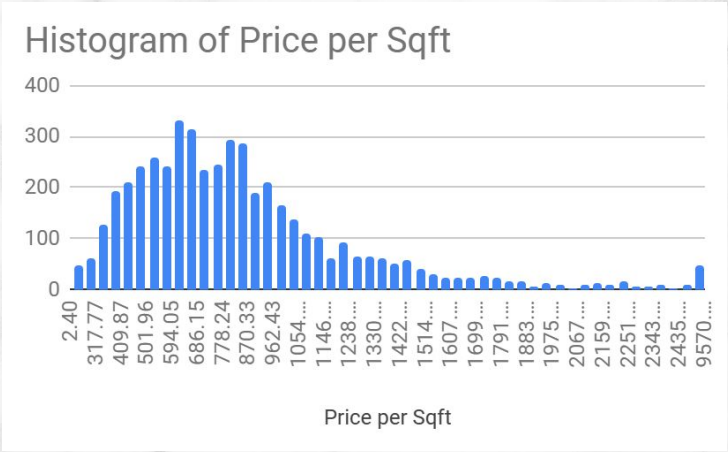
Deskriptif Analytics



Car Parks	
Mean	2
Median	2
Mode	2
Standard Deviation	1
Skewness	4.05
Minimum	1
Maximum	28
Count	3,411

Insight:

1. 1. the data is skewness positively that indicates outlier in + area
2. we need to check the max pint number and see is there really a property with 28 carparks?
3. Thera are gaps (1431) in count data that indicates not all properties have carparks
4. We need to use Median instead of mean for central tendency



Price per Sqft	
Mean	853.12
Median	761.18
Mode	1,000.00
Standard Deviation	474.65
Skewness	4.06
Minimum	2.40
Maximum	9,570.55
Count	4,842.00

Insight:

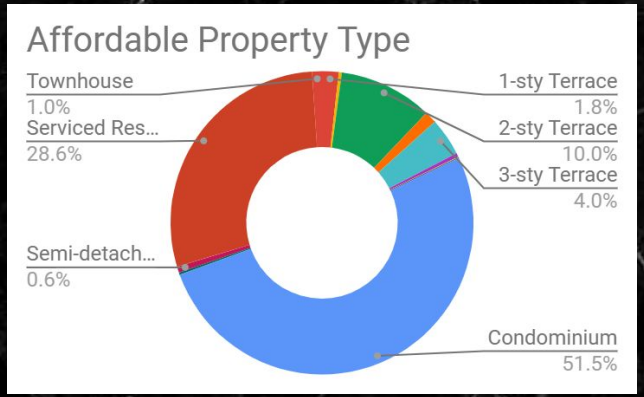
1. The data is skewness positively that indicates outlier in + area
2. We need to use Median instead of mean for central tendency

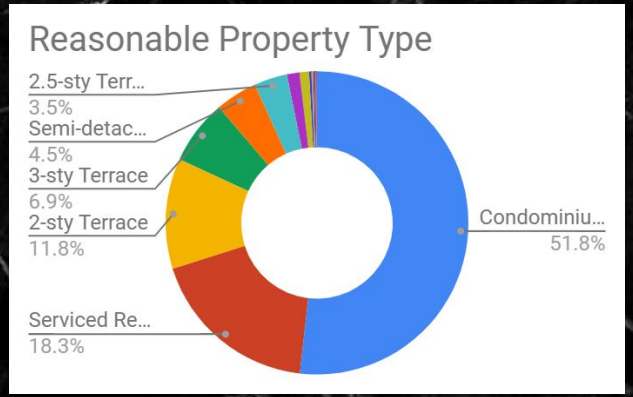
EDA

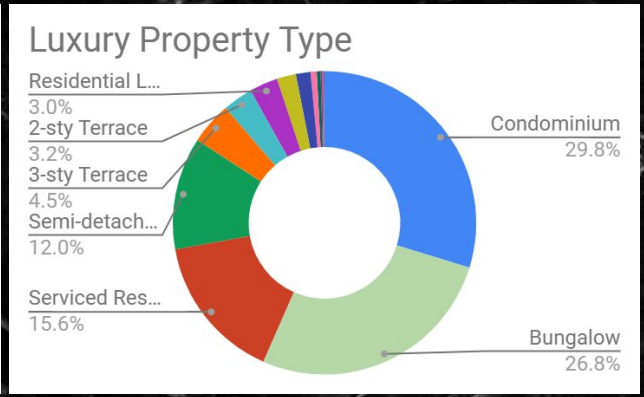
LINK GOOGLE SHEET

Property Class and Its Characteristic

Property Class	Jumlah Property	Avg Price	Median Rooms	Median Bathroom	Median Car parks	Size	Most Property Type	Most Property Character	Most Furnishing Status
Affordable	1,240	1,003,546	3	3	2	1,400	51.5 % Condominium	Build Up	Partly Furnished
Reasonable	1,187	1,809,159	4	4	2	1,900	51.8 % Condominium	Build Up	Partly Furnished
Luxury	1,219	5,323,534	5	5	2	3,979	29.8 % Condominium	Build Up	Partly Furnished

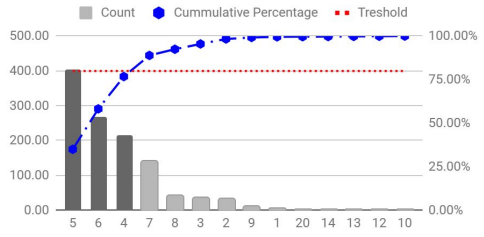






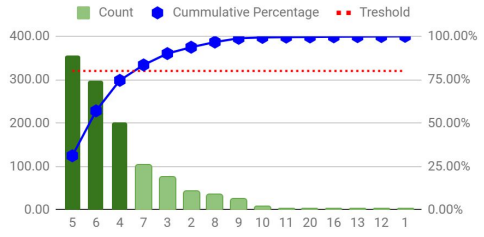
Rooms

Luxury Cass



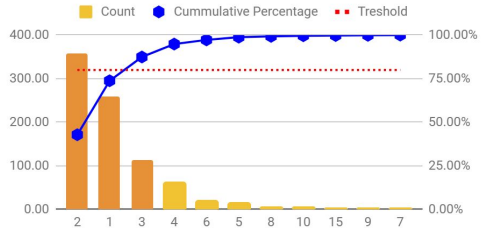
Bathrooms

Luxury Cass

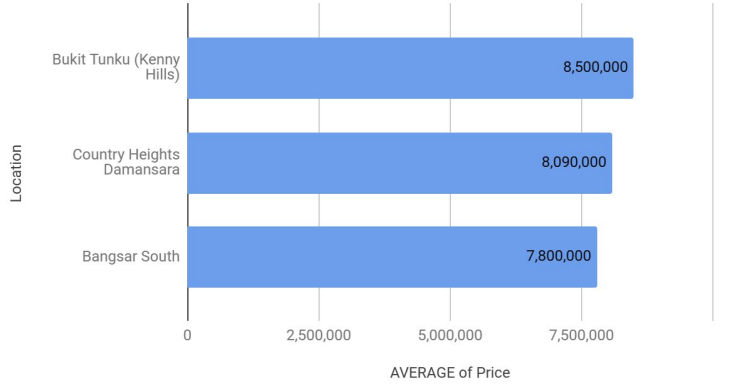


Car Parks

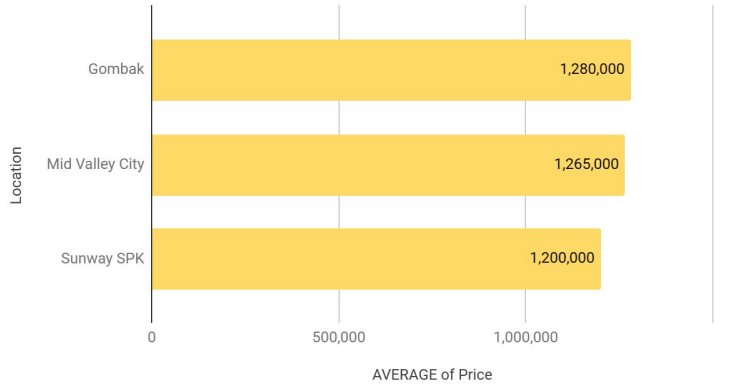
Luxury Class



Price vs. Location

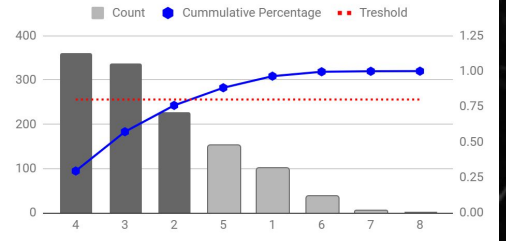


AVERAGE of Price vs. Location (Affordable)



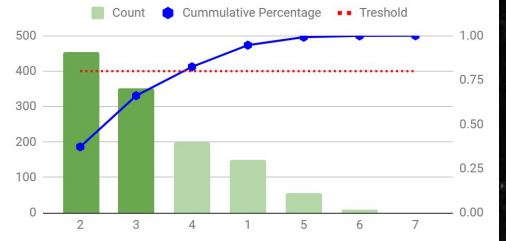
Rooms

Affordable Class



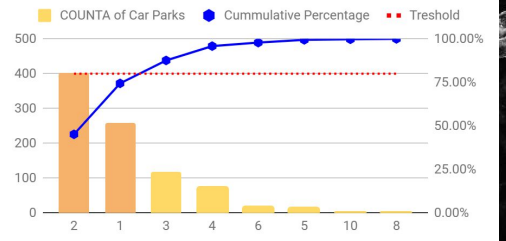
Bathroom

Affordable Class



Car Parks

Affordable Class



Places with high potential profit per property class

Class	Location	COUNTA of Location	MEDIAN of Price	SUM of Price	SUM of Potential Profit
Luxury	KLCC	215	3,500,000	867,889,959	173,577,992
	Mont Kiara	210	3,200,000	849,530,818	169,906,164
	Bangsar	113	6,670,000	778,940,177	155,788,035
	Damansara Heights	106	5,350,000	731,978,898	146,395,780
	Desa ParkCity	126	3,500,000	530,694,350	106,138,870
Reasonable	Mont Kiara	319	1,830,000	591,917,115	118,383,423
	KLCC	210	1,850,000	390,403,831	78,080,766
	Desa ParkCity	150	1,830,000	275,614,000	55,122,800
	Taman Tun Dr Ismail	56	1,630,000	93,082,000	18,616,400
	Bukit Jalil	52	1,770,000	91,004,999	18,201,000
Affordable	KLCC	140	1,050,000	150,404,169	30,080,834
	Mont Kiara	127	1,100,000	134,335,999	26,867,200
	Bukit Jalil	110	928,000	105,213,998	21,042,800
	Dutamas	82	1,035,000	81,422,320	16,284,464
	Cheras	83	968,000	81,211,684	16,242,337

Insight :

- The most property the company offered is a Build up Condominium that is partly furnished
- Each property class has its own characteristic where the price is equivalent to size and fascility.
  - Affordable; 3 rooms, 3 bathroom and 2 carparks
  - Reasonable; 4 rooms, 4 bathroom and 2 carparks
  - Luxury: 5 rooms, 5 bathroom and 2 carparks
- Mont Kiara and KLCC are the areas with the most real estate.
- In along with the price, the most important element affecting profit is the number of properties

Recommendation

- The marketing team can concentrate on Mont Kiara and KLCC, which have the most potential for profit.
- The marketing team should segment the potential customers based on their economic circumstances.
- People can select their desired place depending on their preferences for the property types because most properties are distinct in each region.
- We can recommend Country Heights Damansara, Bukit Tunku and Bangsar South for people who want to live in big house with 6 rooms, 6 Bathroom and 8,681 Sqft property size for RM 1.072/Sqft
- For those looking to reside in a more affordable home with an average 4 rooms, 3 bathrooms, and 2,262 square feet of property size for RM 535.10 per square foot, we can suggest Gombak, Mid Valley City, and Sunway SPK.
- We may suggest properties in KL City, KLCC, and Brickfields with 2 rooms and 2 bathrooms and an overall size of 804 square feet for those looking to live in a home with less rooms.

What to do next? (Based on the recommendation, what are the next plan of action that we can suggest to our company?)

Statistical Measurement

Correlation

LINK GOOGLE SHEET HERE

With Outlier

	Price	*Rooms*	*Bathrooms*	*Car Parks*	*Size*
Price	1				
Rooms	0.7232	1			
Bathrooms	0.7724	0.8241	1		
Car Parks	0.6321	0.6277	0.6477	1	
Size	0.7959	0.6677	0.7106	0.5296	1

Without Outlier

	Price	*Rooms*	*Bathrooms*	*Car Parks*	*Size*
Price	1				
Rooms	0.6916	1			
Bathrooms	0.7470	0.8010	1		
Car Parks	0.6034	0.5630	0.5919	1	
Size	0.7782	0.6603	0.6973	0.4950	1

INSIGHTS

- Size has the highest correlation to price (strong positive correlation)
- Car parks has the lowest correlation to price (low positive correlation)
- There are strong positive correlation between Rooms and Bathrooms
- The lowest Correlation is Size and Car Parks
- Removing outlier make all the correlation in all variable decreased

Comparative Analysis

Regression

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	Model 1
Regression Statistics	
Multiple R	*0.8602*
R Square	0.7399
Adjusted R Square	0.7338
MAPE	17.81%
Standard Error	435,863.0007
Observations	177

	Model 2
Regression Statistics	
Multiple R	0.8582
R Square	0.7365
Adjusted R Square	0.7319
MAPE	17.64%
Standard Error	437,426.4063
Observations	177

	Model 3
Regression Statistics	
Multiple R	0.8423
R Square	0.7095
Adjusted R Square	0.7042
MAPE	17.28%
Standard Error	401697.8334
Observations	169

Notes

For the regression we run three model that in every model has different independent variable that is:

- **MODEL 1** Independent Variable are Rooms, Bathrooms, Car Parks and Size
- We removed Rooms from MODEL 2 since it has a strong corellation with Bathroom (0.824)
- In **MODEL 3** we cut the outlier and make the independent Variable same as **MODEL 2** (bathroom, Size and Car Parks)

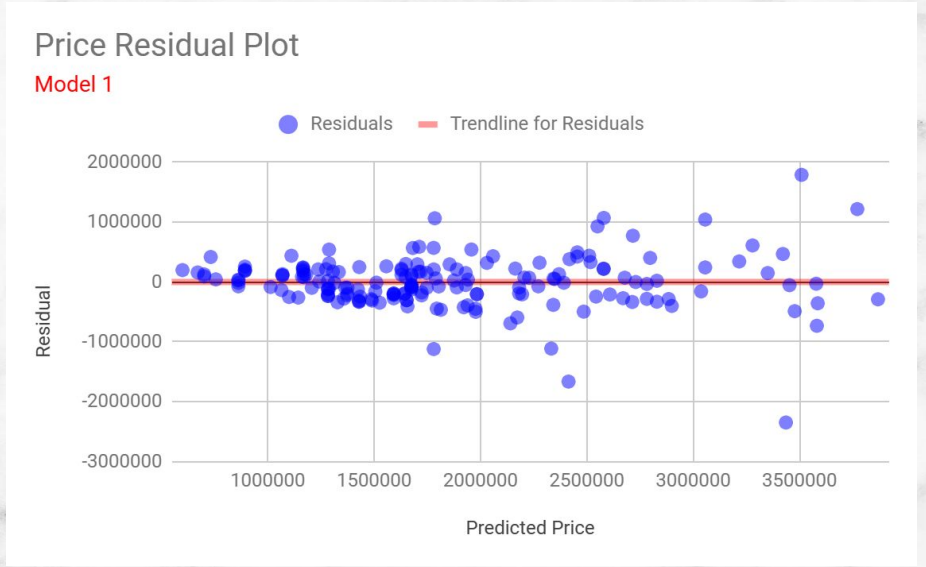
The result is, **MODEL 3** is shown tobe better model, because it has the lowest **MAPE** that is 17.28% and has the lowest **SE**

MODEL 1

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ANOVA					
	df	*SS*	*MS*	*F*	*Significance F*
Regression	*4*	*92942645214849*	*23235661303712*	*122.3080461*	*0*
Residual	172	32675967530914	189976555412		

	Coefficients	*Standard Error*	*t Stat*	*P-value*
Intercept	-211,211	*115,519*	*-2*	*6.92%*
Rooms	71,341	47,632	1	**13.60%**
Bathrooms	151,348	45,060	3	0.10%
Car Parks	160744.5183	54500.31337	2.949423743	0.36%
Size	472.1989033	58.26250062	8.104679653	**0.00%**



INSIGHTS

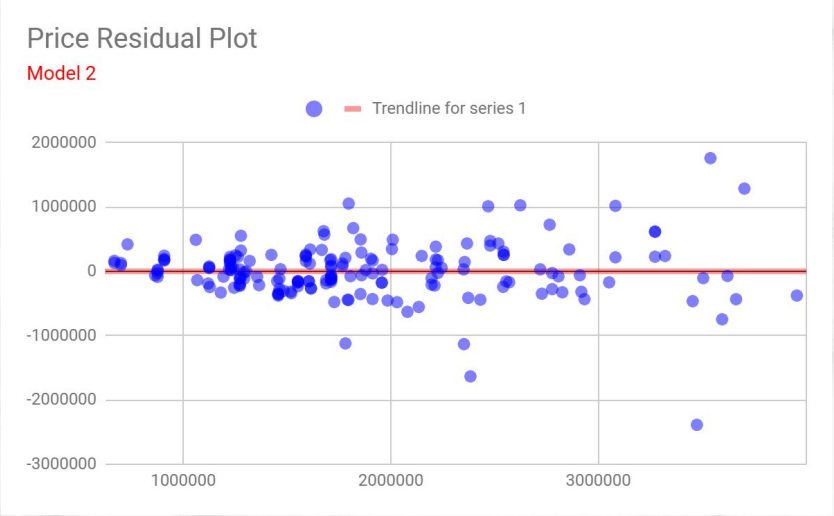
- Significance F less than 5%, so our regression model represents all variables and the model is aceptable
- For this model, we spotted p-value that is more than 5% so we might need to remove it because it means that the independent variable is less significant than others
- The price residual plot shows Heterocedasticity
- We have our model that is -211211+71341*X1+151384*X2+160744.5182*X3+472.1989*X4±e

MODEL 2

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ANOVA					
	df	SS	MS	F	Significance F
Regression	3	92,516,470,807,083	30,838,823,602,361	161	0
Residual	*173*	*33,102,141,938,680*	*191,341,860,917*		

	Coefficients	Standard Error	t Stat	P-value
Intercept	-141,624	106,143	-1	18.39%
Bathrooms	190,889	*36,647*	*5*	*0.00%*
Car Parks	176,807	*53,626*	*3*	*0.12%*
Size	488	57	8	0.00%



INSIGHTS

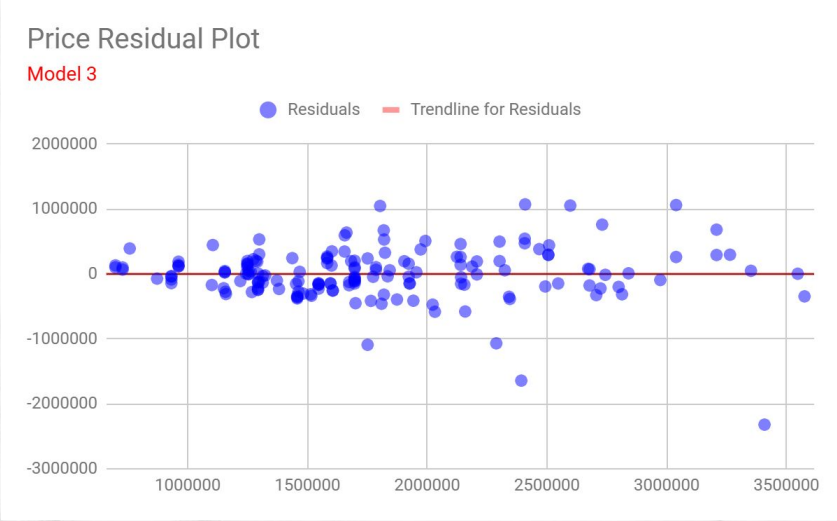
- This model Significance F is less than 5%, so our regression model represents all variables and the model is aceptable as the previous model
- For this model, all p-value is less than 5% so it should perform better and effective in calculating prices
- As the previous model, the price residual plot in this mode also shows Heterocedasticity but a little bit smaller
- We have our model that is -141,624+190889*X1+176807*X2+488.231*X3±e

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MODEL 3

ANOVA					
	df	*SS*	*MS*	*F*	*Significance F*
Regression	3	65029799991389	21676599997130	134.3359296	0
Residual	165	26624589641747	161361149344		

	Coefficients	*Standard Error*	*t Stat*	*P-value*
Intercept	-89733.25907	111307.2042	-0.806176561	42.13%
Bathrooms	168895.6285	35062.21765	4.817026414	0.00%
Car Parks	203724.3525	56814.39205	3.585787776	0.04%
Size	459.2286965	57.69628895	7.959414805	0.00%



INSIGHTS

- This model Significance F is less than 5%, so our regression model represents all variables and the model is aceptable as the other model
- For this model, all p-value is less than 5% and all the p value is smaller in every independent variable compared to other 2 models, so it should perform better and effective in calculating prices
- In this model the Heterocedasticity is smaller in higher predicted prices
- We have our model that is -89733.2590+-89733.2590*X1+-89733.2590*X2+459.2286*X3±e

Price Offer for Customer and Recommendation

Price offer

Estimation for 3 rooms, 4 bathrooms, 3 car parks and 2200Sqft size, the price prediction will be:

Model 3 = -89733.2590 + -89733.2590*3 + -89733.2590*4 + 459.2286*2200 ± 401697.8334

Min: **RM 1,805,628**

Max: **RM 2,609,023**

Recommendation

- We must acknowledge that our regression model deviates from the homoscedasticity assumption, thus we must speak with our head of data to determine our next path of action.
- We need need to review, consult and remove outlier to make better regression model
- We need to try to various combination of independent variable to make a good model
- We need to check wheather our model fullfill all of the assumption of linear regression



THANK YOU

Akhmad Fauzi Nurulhamzah